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The Archivist

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The Archivist, as an agent of retention and recollection [1], navigates retention strata to excavate old memories, expressed through the degradation that antiquated media impart.[1] The instrument builds on the memorial context and aesthetic of existing devices—including the Chase Bliss MOOD [2], Fairfield Circuitry's Shallow Water [3], four-track cassette (e.g., Tascam Portastudio) and tape splicing—extending these with the CV capabilities of Eurorack (e.g., [4], [5], [6]).

Both modes read a recording x in overlapping units, unit j starting at time t_j from position a_j at pitch p:

$$y(t) = \sum_j g_j(t)\, x\big(a_j + p\,(t - t_j)\big). \tag{1}$$

In granular synthesis (GRAINS), units are grains spaced d/N apart, with $g_j(t) = g\, w\big((t - t_j)/d\big)$ for gain g and a Hann-to-*exp* window w of duration $d \in [0.05, 0.5]$ s, and $a_j = s\, t_j + \sigma\xi_j + \delta_j$ for speed $s \in [-2, 2]$ with optional position offset, Gaussian spray $\sigma\xi_j$ and alignment shift δ_j. Fill factor [7], spawn rate × duration, is the mean overlap $N = N_0 \max(1, |s/p|)\,\Gamma(\sigma) \le 40$, where density control sets base overlap $N_0 \in [0.25, 4]$, with synchronous or stochastic grain timing; the second factor keeps the source covered; and $\Gamma \ge 1$ extends density with the ataxy $u \in [0, 1]$ [8]. Spray grows with u from a quarter of the drift $(d/N)\,|s - p|$ between successive grains, and g passes from $1/N$ to $1/\sqrt{N}$ as grains decorrelate. At $s = p$ and $\sigma = 0$ on a single stratum every grain reads $x(pt)$, so synchronous Hann grains at integer $N \ge 2$ reconstruct it; for moderate $s \ne p$ a correlation search (after WSOLA [9]; coarse alignment [4]) sets δ_j for pitch and time scaling, and increased ataxy produces clouds. While these are generally offered as separate modes, The Archivist provides access to the continuum of possible states with simultaneous speed and position control over the read head.

TAPE emulates the Eltro Mark II Rate Changer [10], one of the first devices capable of separating pitch and time. Units are passes of the four heads at 90° on its rotating drum, spaced $T = A/|v|$ apart for head spacing A and head velocity $v = p - r$ against tape speed r, with $a_j = r\, t_j - \max(v, 0)\, T + \epsilon_j$; g_j is a crossfade $\chi(q) = (3q^2 - 2q^3)^{0.7}$ scaled by each head's level, and ϵ_j its placement error. The loop has length $W \in [B/64, B]$ for a bar B, and $A = B/n$, $n \in \{1, 2, 4, \ldots, 32\}$; with artistic licence, A is scaled by $|v|$ when $r \ne 1$ so that each pass lasts B/n and crossfaded splices fall on note values. Tape echo follows tape loop length with print-through modelling via feedback delay of one loop period with per-pass treble loss. Disintegration integrates wear η on a capture, $\mathrm{d}\eta/\mathrm{d}t = c^{0.48}\,|p|/(300\,\mathrm{s})$ for wear setting c, driving level loss, treble loss and dropouts. k-field modulation, after [3], low-passes noise (0.8–15 Hz) onto a delay tap (±12 ms limit).

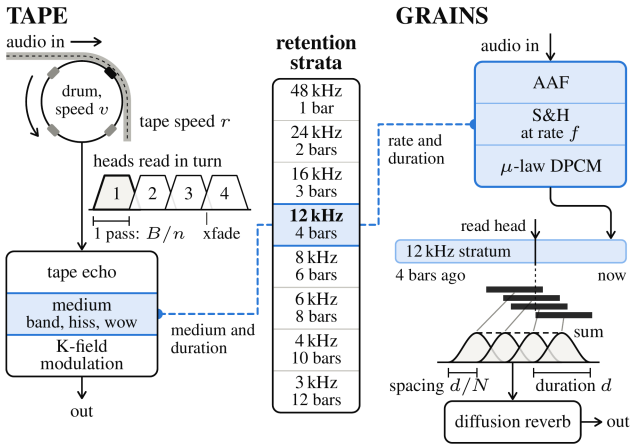


Figure 1: The Archivist TAPE and GRAINS modes and retention strata.

Eight strata pair rates from 48 kHz down to 3 kHz with spans of 1–12 bars (Fig. 1). GRAINS records all eight at once. Below 48 kHz each is stored through a model of the E-mu EII codec (1984): after a low-pass and sample-and-hold to the stratum rate f, each sample z_m is coded by 8-bit DPCM with a μ-law companded residual [**?**],

$$x_m = x_{m-1} + D\,Q\left(\frac{z_m - x_{m-1}}{D}\right), \tag{2}$$

where Q is the μ-255 quantiser, on the EII table, clipping at ± 1, and $D = 0.15 + 0.075 \log_2(48\,\mathrm{kHz}/f)$. Because $|x_m - x_{m-1}| \le D$, a sinusoid of amplitude α is tracked only up to $Df/(2\pi\alpha)$; above that the coder slope-overloads, a level-dependent dulling that deepens as strata coarsen. TAPE imposes matched band limits, hiss, wow and dropouts at playback. Selection steps between strata or blends neighbours:

[1] github.com/jhockman/the-archivist

on a capture this mixes coarser, longer strata into the present; uncaptured, it colours the live signal.

I. References

[1] E. Husserl, *On the Phenomenology of the Consciousness of Internal Time (1893–1917)*. Kluwer, 1991, trans. J. B. Brough.

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[3] Fairfield Circuitry, "Shallow Water: K-field modulator," spec. sheet.

[4] É. Gillet, "Clouds (2015) and Beads (2021): Manuals and firmware source," Mutable Instruments.

[5] Make Noise, "Morphagene manual," v2.4.

[6] Instruō, "arbhar user manual," firmware 2.0.

[7] C. Roads, *Microsound*. MIT Press, 2001.

[8] I. Xenakis, *Formalized Music*, rev. ed. Pendragon Press, 1992.

[9] W. Verhelst and M. Roelands, "An overlap-add technique based on waveform similarity (WSOLA) for high quality time-scale modification of speech," in *Proc. IEEE ICASSP*, 1993, pp. 554–557.

[10] W. Carlos, "The Eltro and the voice of HAL," 2008.